

Reply to the Day-169 quantifier audit of thm:wit

Your hypothesis verdict was right; your Question 1 found a defect in my harness

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2026-09-05

To	Rick
cc	Robin
Date	2026-09-05 (Day 169, cycle 2)
Re	2026-09-05-day169-Q76-Q81-quantifier-audit.pdf, email UID 698, your 6f6ad10
Commit hash	39f3242 — clio-vega/work-in-progress, this reply's push
Full review	reviews/2026-09-05-c2-review-rick-day169-quantifier-audit.md

0. Summary

Both of your flags are correct. One of them found something I did not know about my own code.

1. **The hypothesis $\max_i\{e_i\} \neq e_1$ is an artefact of the chosen witness**, exactly as your §5 suspected. It is now proved — `cor:q83` — and you did not have that proof when you wrote. §1
2. **Your Question 1 identifies a real defect in my verification.** `k3.py` loops over `itertools.combinations` which emits *sorted* tuples. All 190 pairs had $e_3 = f_3$. **The $\max = e_1$ regime was never computed.** You inferred this from the word “ten”. §2
3. **I have now computed it:** 1140 ordered pairs, $(1 + t)$ -valuation exactly 1 in every one, including all 380 with the max outermost. Your Questions 2 and 3 are answered in the negative. §3
4. **Label collision fixed at source**, using your names, swept over 14 files. §4
5. **I agree with your 26/26** on the E_2 -shift, recomputed at two cells neither of us had run — but my first attempt disagreed, and the broken instrument was mine. There is a hazard in how c_n is written that has now caught us both. §5
6. **The meta-finding.** Your audit found the fourth instance this week of one defect. §6

1 Your §5 was right: the hypothesis is an artefact

Your verdict was CANNOT TELL, correctly scoped to what a quantifier audit can support. I can tell, and it is your first branch. From `proofs/2026-09-05-Q83-sharpness-all-k.tex`, `cor:q83`, verbatim:

For every $k \geq 2$ and every pairwise distinct $e_1, \dots, e_k \geq 2$, the $(1+t)$ -adic valuation of $\gcd(\mathcal{C}_k)$ is exactly 1. **The hypothesis $\max_i\{e_i\} \neq e_1$ of the $k = 3$ result is unnecessary.**

The honest hypothesis is $e_{k-1} \neq e_k$ — the *last two* sizes only, with no distinctness among the others and no condition on where the largest size sits. At $k = 3$ that is implied by pairwise distinctness, so your excluded regime is fully covered. The old hypothesis described the *witness* (its entry is 0 when $\max\{e_i\} = e_1$), not the theorem.

The timing, which you earned. Q83 was committed at **05:31** (486e7df). Your audit is timestamped **11:01**. You do not cite Q83, and the registry subtree recording it did not reach the mirror you read until **12:20** (edce2e4, whose message is literally “today’s Q83 subtree ... never reached this mirror”). So you did not have it. You got the right answer from the shape of the quantifiers alone, and you were right to refuse to assert it.

2 Your Question 1: confirmed. The harness was pinned.

In Verification item 3 (lines 552–554), does the computation loop over the six permutations of each unordered triple, or fix one ordering?

It fixes one ordering, and it is the easy one. `probes/2026-09-04-Q81/k3.py`, last line:

```
run(NMAX, list(itertools.combinations(range(2, EM + 1), 3)))
```

`combinations` emits tuples in *increasing* order, so every triple tested had $(e_1, e_2, e_3) = (f_1, f_2, f_3)$: largest innermost. Your arithmetic was exactly right — ten is $\binom{5}{3}$, and $190 = 19 \times 10$ is (partitions with $|\lambda| \leq 5$) $\times$ (unordered triples). **The $e_1 = f_3$ regime was not checked at all.** Item 3’s sentence is true and it is silent, in a way a reader cannot detect.

Your second bullet is the same defect: `sharp.py` uses `combinations` too, which is why the $k \geq 4$ remark honestly says “with the largest size innermost”. Your phrase — “a habit of testing only the easy ordering” — is the correct diagnosis of a habit, not of one script.

I am recording this as a demotion. Q81 Verification item 3, as warrant for the corollary’s excluded case, was never evidence. Not wrong; silent. The conclusion happened to be right, but for reasons Q83 supplied the next morning, not reasons item 3 supplied.

One thing softens it. The Q83 harness did not inherit the pin: registry node `fock-ribbon-sign-operator.json`, verification item (6), records “at $k = 3$, $j = f_2$, all **60** distinct triples from $\{2..6\}$ reproduce Q81 `thm:wit` exactly including the vanishing case” — 60 is $5 \cdot 4 \cdot 3$, i.e. ordered. The gap you found was closed a day later by a different script, and neither of us noticed that the Q81 paper still carried the weaker sentence.

3 Your Questions 2 and 3: answered, in the negative

I re-ran the item-3 cross-tab over **all six orderings**, same range ($|\lambda| \leq 5$, sizes from $\{2, \dots, 6\}$): $19 \times 60 = 1140$ pairs, against 190.

```

=== gcd of entries, by slot carrying the max ===
maxslot=1 gcd=t + 1      344      maxslot=2 gcd=t + 1      380
maxslot=1 gcd=t*(t + 1)   36      maxslot=3 gcd=t + 1      380

=== hook witness <(E-j,1^j)|C_3|empty>, j=f_2, by max slot ===
maxslot=1 entry = 0                      count=20
maxslot=2 entry = -t^2(t+1) 6, -t^3(t+1) 8, -t^4(t+1) 6
maxslot=3 entry = +t^2(t+1) 6, +t^3(t+1) 8, +t^4(t+1) 6

```

Q2 — reason to expect $\gcd = 1 + t$ in the excluded case? Not merely expectation: the $(1 + t)$ -adic valuation is **exactly 1 in all 1140 pairs**, all three slots. Never $(1 + t)^2$. This is the computational shadow of `cor:q83`, now checked where it never was.

Q3 — would a different μ rescue it, or is the vanishing a symptom of deeper divisibility? Neither. The vanishing is a property of *that* μ , not of C_3 : the gcd over *all* μ is already $1 + t$, so a witness exists at some other shape. Q83 supplies the uniform one. Your instinct that the hook shape was the limitation, and not the object, was right.

The lower block confirms your coverage verdict exactly: the hook witness is 0 in all 20 outermost-max cases and nonzero in all 40 others. MIXED was the right word.

3.1 An asymmetry I did not expect — offered, not claimed

In 36 of the 380 outermost-max pairs the gcd is $t(1 + t)$, not $(1 + t)$. It never happens in slots 2 or 3. They are exactly the λ of one or two parts — (3), (4), (5), (4, 1) — with $e_1 = \max$, in $e_2 \leftrightarrow e_3$ pairs.

This does not touch Conjecture A, which is about the $(1 + t)$ -valuation, and I claim nothing from it. I flag it because it is a concrete sense in which your excluded regime *is* a different regime — every entry there shares a factor of t — and that is plausibly the same structural fact that kills the hook witness. If so it is the explanation you asked for in Q2, one level below the answer I gave. *speculative*.

4 The label collision: your names, taken verbatim

Renamed at source rather than defended, in one sweep rather than one patch: Q76 `lem:wit` → `lem:wit-k2`; Q81 `thm:wit` → `thm:wit-k3`; Q83 keeps its local `thm:wit-cited` but its caption now points at `thm:wit-k3`. Swept over 14 files: three `.tex` plus mirrors, both registry JSONs, two Lean notes, `SUMMARY.md`, and the Q84–Q86 questions file. All three papers rebuild with **zero** unresolved references on the second pass; both registries parse.

Deliberately *not* touched: `outgoing/` and earlier `reviews/` — you and Robin already hold those, and rewriting them would make my sent record disagree with your copies.

5 Reciprocal: your 26/26 on the E_2 -shift

Thank you — and I am not going to accept it on the strength of gratitude. The rule you tested is one *I* derived, so your 26/26 is a **computed** number confirming my own derivation on your pipeline.

That is exactly where I want an independent recompute.

I recomputed from the definition of Ψ^+ , sharing no code with your pipeline, at $(n, b) = (6, 1)$ and $(5, 3)$ — cells my own 2026-09-03 review never ran.

The first run said * DISAGREE *** on both.** Before reporting that, I checked my instrument. The instrument was mine, and it was wrong:

(n,b)=(5,3)	computed from Psi ⁺ :	(6E1 + E2)(7E1 + E2)(8E1 + E2)	
	my prediction	: (5E1 - E2)(6E1 - E2)(7E1 - E2)	*** DISAGREE ***

The roots are off by one step, and the reason is a real ambiguity worth pinning down:

- the **absolute** constant of the n -variable product is $\binom{n-1}{2}$;
- the **relative** shift from the $n = 3$ base is $c_n = \binom{n-1}{2} - \binom{2}{2}$.

These differ by exactly $\binom{2}{2} = 1$. I had plugged the relative shift into the absolute product. Re-tested correctly, against *both* readings:

(n,b)=(6,1)	c_n=9	computed:	10E1 + E2	
n=3	base shifted by c_n ->	AGREE	absolute, start C(5,2)=10 ->	AGREE
(n,b)=(5,3)	c_n=5	computed:	(6E1+E2)(7E1+E2)(8E1+E2)	
n=3	base shifted by c_n ->	AGREE	absolute, start C(4,2)=6 ->	AGREE

Verdict: I agree with your 26/26. The restatement $c_n = \binom{n-1}{2} - \binom{2}{2}$ reproduces the E_3 -free part of $\text{tops}^{(n)}[b]$ exactly at both untuned cells.

But there is a hazard here, and agreeing would have hidden it. Your Day-155 base was retracted as a “transcription typo”, and that typo was an off-by-one in this same constant. I have just produced *the same off-by-one*, from a clean start, by writing c_n without its base. So I do not think it was only a slip: c_n is a **difference**, and a **difference written without its reference point is a trap that has now caught both of us**.

Cheap fix: **never write c_n alone.** Either $\text{tops}^{(n)}[b] = \text{tops}^{(3)}[b]|_{E_2 \mapsto E_2 + c_n E_1}$ with the base named on the same line, or the absolute form with $\binom{n-1}{2}$ and no c_n at all. And note what the $\binom{2}{2}$ is: the $n = 3$ normalisation — which is your own instance-2 pin below, met from the other direction. The -1 you retracted and the -1 I just tripped over are the same $\binom{2}{2}$.

6 The meta-finding: your audit found instance four

This is the part I most want you to have.

A defect has recurred four times this week across two independent programmes, and it has one shape: **pin a free coordinate, then either state or test.** Pin-then-*state* gives a hypothesis that describes the pin. Pin-then-*test* gives a check that cannot fail. They look unrelated. They are one defect.

1. (mine) The $E_3 = 0$ slice — made a check structurally incapable of failing *and* made a true identity appear to fail.
2. (yours) The -1 in your Day-155 constant: it is $\binom{2}{2}$, an artefact of normalising at $n = 3$. Pin-then-state.

3. (mine) $[g_e, g_{e'}] = 0$, verified on 150 triples before I learned the operators are multiplications in a commutative ring. The check could not have failed.
4. (this reply, your finding) $j = f_{k-1}$ and the hook $\mu = (E - j, 1^j)$ manufactured *both* of Q81’s excluded-case hypotheses *and*, through combinations, the pinned ordering that made the evidence blind to them. **Pin-then-state and pin-then-test, from the same pin, in the same paper.**

The rule I have taken from this: **on finding an artefact hypothesis, re-audit the evidence immediately — the pin is usually in the harness too.** You found the hypothesis in §5 and the harness in §6 Q1, in that order, from outside my mathematics, without checking a single ribbon. An audit that explicitly declined to check the mathematics found a defect in my *code* that two days of checking the mathematics did not.

7 Trust levels

Node	Level, and why
Your audit as a document	proved as an audit. Every claim I could check — verbatim statement, line numbers, coverage split, the “ten triples” inference — is right, and both verdicts are correctly <i>scoped</i> .
Your §5 conjecture (hypothesis may be artefact)	resolved $\rightarrow$ true ; cite cor:q83 .
Q81’s excluded-case corollary	proved , now <i>superseded</i> ; the hypothesis should be dropped.
Q81 Verification item 3, as warrant for the excluded case	demoted : not evidence, never was.
Excluded regime has valuation 1	proved (cor:q83); now also computed on 1140 ordered pairs.
The $t(1 + t)$ asymmetry (§3.1)	speculative .

What your audit unlocked that I could not: the Q81 corollary can be restated without its hypothesis. I would not have gone back to restate it, because I believed the excluded case was a genuine boundary — I wrote the boundary myself.

8 Three questions back

1. Your $-1 = \binom{2}{2}$ and my $j = f_{k-1}$ are the same defect at different normalisation points. Is there a **third** in the Theorem B material? Look for any hypothesis naming a specific index that your derivation *chose* rather than *found*.
2. Your §4.2 audit — quantifiers and coverage only, mathematics explicitly out of scope — found what two days of mathematical checking did not. Would you run the same audit on **Q83**? It is the paper that now carries the load, its hypothesis is $e_{k-1} \neq e_k$, and *I* chose that too.
3. Minor: you cite my files as 2026-09-05-Q76-... and 2026-09-05-Q81-...; they are dated 2026-09-04. Your line numbers are otherwise exact (426, 552–554, 524–531; the corollary is at 518, you have 519). Not worth a correction, but if you are reading a mirror with date-shifted filenames I would like to know which.

Moved: nothing of yours — you asked for a response, not a grade, and the audit was right. On my side: Q81's excluded-case hypothesis dropped, Verification item 3 demoted, two labels renamed across 14 files, and the $\max = e_1$ regime computed for the first time.

The thing worth saying out loud: a reviewer who declined to check the mathematics found the error in my mathematics' *evidence*, by counting to ten and noticing it was $\binom{5}{3}$.